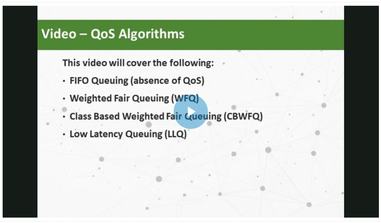
Queuing Algorithms

9.3.1

Video Tutorial - QoS Algorithms

Click Play for an overview of the different types of QoS queuing algorithms.



9.3.2

Queuing Overview

The previous topic covered traffic characteristics. This topic will explain the queuing algorithms used to implement QoS. The QoS policy implemented by the network administrator becomes active when congestion occurs on the link. Queuing is a congestion management tool that can buffer, prioritize, and, if required, reorder packets before being transmitted to the destination.

A number of queuing algorithms are available. For the purposes of this course, we will focus on the following:

- First-In, First-Out (FIFO)
- Weighted Fair Queuing (WFQ)
- Class-Based Weighted Fair Queuing (CBWFQ)
- Low Latency Queuing (LLQ)

9.3.3

First In First Out

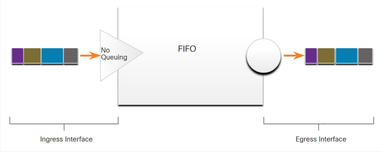
In its simplest form, First In First Out (FIFO) queuing, also known as first-come, first-served queuing, buffers and forwards packets in the order of their arrival.

FIFO has no concept of priority or classes of traffic and consequently, makes no decision about packet priority. There is only one queue, and all packets are treated equally. Packets are sent out an interface in the order in which they arrive, as shown in the figure. Although some traffic may be more important or time-sensitive based on the priority classification, notice that the traffic is sent out in the order it is received.

When FIFO is used, important or time-sensitive traffic can be dropped when there is congestion on the router or switch interface. When no other queuing strategies are configured, all interfaces, except serial interfaces at E1 (2.048 Mbps) and below, use FIFO by default. (Serial interfaces at E1 and below use WFQ by default.)

FIFO, which is the fastest method of queuing, is effective for large links that have little delay and minimal congestion. If your link has very little congestion, FIFO queuing may be the only queuing you need to use.

FIFO Queuing Example

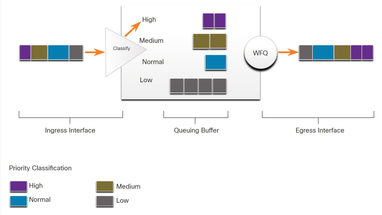


9.3.4

Weighted Fair Queuing (WFQ)

Weighted Fair Queuing (WFQ) is an automated scheduling method that provides fair bandwidth allocation to all network traffic. WFQ does not allow classification options to be configured. WFQ applies priority, or weights, to identified traffic and classifies it into conversations or flows, as shown in the figure.

Weighted Fair Queuing Example



WFQ then determines how much bandwidth each flow is allowed relative to other flows. The flow-based algorithm used by WFQ simultaneously schedules interactive traffic to the front of a queue to reduce response time. It then fairly shares the remaining bandwidth among high-bandwidth flows. WFQ allows you to give low-volume, interactive traffic, such as Telnet sessions and voice, priority over high-volume traffic, such as FTP sessions. When multiple file transfers flows are occurring simultaneously, the transfers are given comparable bandwidth.

WFQ classifies traffic into different flows based on packet header addressing, including such characteristics as source and destination IP addresses, MAC addresses, port numbers, protocol, and Type of Service (ToS) value. The ToS value in the IP header can be used to classify traffic.

Low-bandwidth traffic flows, which comprise the majority of traffic, receive preferential service which allows their entire offered loads to be sent in a timely fashion. High-volume traffic flows share the remaining capacity proportionally among themselves.

Limitations

WFQ is not supported with tunneling and encryption because these features modify the packet content information required by WFQ for classification.

Although WFQ automatically adapts to changing network traffic conditions, it does not offer the degree of precise control over bandwidth allocation that CBWFQ offers.

9.3.5

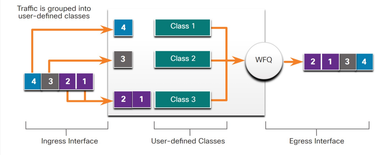
Class-Based Weighted Fair Queuing (CBWFQ)

Class-Based Weighted Fair Queuing (CBWFQ) extends the standard WFQ functionality to provide support for user-defined traffic classes. With CBWFQ, you define traffic classes based on match criteria including protocols, access control lists (ACLs), and input interfaces. Packets satisfying the match criteria for a class constitute the traffic for that class. A FIFO queue is reserved for each class, and traffic belonging to a class is directed to the queue for that class, as shown in the figure.

When a class has been defined according to its match criteria, you can assign it characteristics. To characterize a class, you assign it bandwidth, weight, and maximum packet limit. The bandwidth assigned to a class is the guaranteed bandwidth delivered to the class during congestion.

To characterize a class, you also specify the queue limit for that class, which is the maximum number of packets allowed to accumulate in the queue for the class. Packets belonging to a class are subject to the bandwidth and queue limits that characterize the class.

CBWFQ Example



After a queue has reached its configured queue limit, adding more packets to the class causes tail drop or packet drop to take effect, depending on how class policy is configured. Tail drop means a router simply discards any packet that arrives at the tail end of a queue that has completely used up its packet-holding resources. This is the default queuing response to congestion. Tail drop treats all traffic equally and does not differentiate between classes of service.

9.3.6

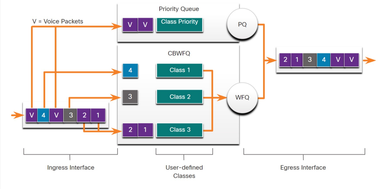
Low Latency Queuing (LLQ)

The Low Latency Queuing (LLQ) feature brings strict priority queuing (PQ) to CBWFQ. Strict PQ allows delay-sensitive packets such as voice to be sent before packets in other queues. LLQ provides strict priority queuing for CBWFQ, reducing jitter in voice conversations, as shown in the figure.

Without LLQ, CBWFQ provides WFQ based on defined classes with no strict priority queue available for real-time traffic. The weight for a packet belonging to a specific class is derived from the bandwidth you assigned to the class when you configured it. Therefore, the bandwidth assigned to the packets of a class determines the order in which packets are sent. All packets are serviced fairly based on weight; no class of packets may be granted strict priority. This scheme poses problems for voice traffic that is largely intolerant of delay, especially variation in delay. For voice traffic, variations in delay introduce irregularities of transmission manifesting as jitter in the heard conversation.

LLQ allows delay-sensitive packets such as voice to be sent first (before packets in other queues), giving delay-sensitive packets preferential treatment over other traffic. Although it is possible to classify various types of real-time traffic to the strict priority queue, Cisco recommends that only voice traffic be directed to the priority queue.

LLQ Example



9.3.7

Check Your Understanding - Queuing Algorithms

Check your understanding of queuing algorithms by choosing the BEST answer to the following questions.

1. Which queuing algorithm simultaneously schedules interactive traffic to the front of a queue to reduce response time?
   - FIFO
   - WFQ
   - CBWFQ
   - LLQ

2. Which queuing algorithm provides support for user-defined traffic classes?
   - FIFO
   - WFQ
   - CBWFQ
   - LLQ

3. Which queuing algorithm is effective for large links that have little delay and minimal congestion?
   - FIFO
   - WFQ
   - CBWFQ
   - LLQ

4. Which queuing algorithm classifies traffic into different flows based on packet header addressing?
   - FIFO
   - WFQ
   - CBWFQ
   - LLQ

5. Which queuing algorithm allows delay-sensitive data such as voice to be sent before packets in other queues?
   - FIFO
   - WFQ
   - CBWFQ
   - LLQ

6. Which queuing algorithm applies priority, or weights, to identify traffic and classify it?
   - FIFO
   - WFQ
   - CBWFQ
   - LLQ

Check

Show Me

Reset